

Technical Document #863: Database Systems - Comprehensive Overview

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Database replication copies data across multiple servers. It improves availability and provides fault tolerance.

Sharding distributes data across multiple database servers. It improves performance by parallelizing query processing.

Database monitoring tracks performance metrics and identifies bottlenecks. Query analysis and indexing strategies optimize database performance.

Database migration tools manage schema changes and data transformations. Flyway and Liquibase are popular database migration tools.

Database indexing improves query performance by creating data structures that allow faster data retrieval.

NoSQL databases handle unstructured and semi-structured data. Document stores, key-value stores, and graph databases are common NoSQL types.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Key Takeaways:

- Database connection pooling reuses database connections to reduce overhead.
- Database caching stores frequently accessed data in memory.
- Sharding distributes data across multiple database servers.